Spatial–Temporal Feature Contributions and Model Reliance in New York City Taxi Travel-Time Estimation

Le Phuc An, Phu Van Tho, Pham Khanh, Thu Le, and Nhat-Tung Le
FPT University, Ho Chi Minh City, Vietnam
Corresponding author: tungln4@fpt.edu.vn

***Abstract*—We compare spatial/routing descriptors with calendar encodings and completed-trip history for taxi travel-time estimation when pickup time and endpoint coordinates are known at dispatch. We examine both the gains from adding these features and their use by fitted models, then compare learners as training-set size increases. Using 1,110,000 chronologically split 2015 NYC Yellow Taxi trips, we compare LightGBM, CatBoost, RealMLP and TabM under hyperparameters selected once on the full feature set and then held fixed. With 90,000 training trips, time/history features (F2) yield test MAE 33.73–47.95 s below spatial/routing features (F1); all Bonferroni-adjusted pickup-day bootstrap intervals exclude zero. Combining both bundles (F3) gives the lowest mean MAE for each learner (200.14–204.90 s), below TEMP (255.97 s) and uncalibrated OSRM durations (527.31 s). On October diagnostic trips, permuting OSRM descriptors in fitted F3 models produces the largest penalty (252.02–301.36 s), measuring reliance rather than refitted marginal value. Across seven nested training tiers from 90k to 1.08M trips, tree MAE varies within 199.50–203.39 s. RealMLP reaches 192.72 s at 720k and 192.97 s at 900k, but rises to 200.08 s at 1.08M because one of three seeds degrades sharply. The scaling results are exploratory because training size, update count and history density change together and tiers were compared on test data.**

Index Terms—taxi travel time; feature ablation; model reliance; sample scaling; tabular deep learning

I. Introduction

Choosing inputs for a travel-time model requires measuring how much they improve predictions after refitting. Attribution in a full-feature model answers a different question: a feature may receive a large attribution even when adding it produces only a small gain [1]–[4]. We compare these two measurements for taxi trips. We also compare boosted trees and neural models across training sizes, since their relative performance varies with the dataset and representation [14], [15].

Our comparison starts from pickup time and endpoint coordinates. We add spatial/routing features (F1), a time/history bundle (F2), or both (F3). Only the Open Source Routing Machine (OSRM) descriptors require an external routing query [6]; zones and geometric features are derived from coordinates. F2 includes both calendar encodings and completed-trip duration history; we evaluate their combined contribution. We evaluate LightGBM [10], CatBoost [11], RealMLP [12] and TabM [13] on a chronological cohort of 1,110,000 NYC Yellow Taxi trips from 2015. Our contributions are:

1) A same-trip comparison of the base (F0), spatial/routing (F1), time/history (F2) and full (F3) sets across four learners and three seeds on 90,000 training trips, with paired pickup-day bootstrap intervals and one hyperparameter setting per learner selected on F3.
2) A comparison of refitted feature gains with grouped permutation and permutation-SHAP reliance on the same full-feature models, using October diagnostic trips.
3) A sample-scaling analysis across seven nested tiers from 90k to 1.08M training trips, with history encodings recomputed within each tier.

II. Related Work

A. Route and historical information for travel time

Wang et al. [5] estimate travel time from neighboring trips and adjust historical durations using a temporal speed reference. We use an adapted deterministic reference, denoted TEMP. OSRM supplies static road-network route descriptors [6]. DeepTTE [7] instead uses observed trajectories, which are unavailable at dispatch. Huang et al. [8] study routing descriptors with tree ensembles, and Schleibaum et al. [9] study route-free estimation with stacked ensembles. Differences in cohorts, filtering and evaluation prevent direct numerical comparisons with our results.

B. Feature ablation, model reliance and sample scaling

LightGBM and CatBoost are boosted-tree learners; RealMLP and TabM are neural tabular learners [10]–[13]. Benchmarks compare their model families across tabular datasets [14], [15]. Cherepanova et al. [16] assess feature selection through downstream accuracy, whereas we compare fixed feature bundles. Here, an ablation refits a learner after changing its feature set. Grouped permutation [1], [2] and SHAP [3] instead examine fixed models. Permuting correlated features can create implausible inputs [4]; a permutation penalty therefore need not equal the benefit of acquiring those features and refitting.

III. Data and Feature Construction

A. Chronological cohort and nested training tiers

We use the 2015 Microsoft Azure NYC TLC Yellow Taxi source [17]. Before sampling, eligibility checks require ordered pickup and drop-off times, duration of 1–180 min, finite coordinates strictly inside valid TLC taxi-zone polygons, passenger count at least one, and rate codes 1–2. Recorded-distance and Haversine speeds must not exceed 96.56 km/h (60 mph). We exclude zero-distance trips longer than 2 min and trips crossing the September–October, October–November or December–January boundaries. Realized taximeter distance is used for filtering and excluded from all predictors. These

Table I
CHRONOLOGICAL PARTITIONS AND NESTED TRAINING TIERS.

Partition	Months	Per month	Trips
TRAIN (S90k)	Jan–Sep	10,000	90,000
TRAIN (larger tiers)	Jan–Sep	20k–120k	180k–1,080k
DIAG	Oct	10,000	10,000
TEST	Nov–Dec	10,000	20,000

retrospective eligibility checks limit interpretation as a fully prospective dispatch evaluation.

The January–September pool contains 1,080,000 trips, or 120,000 per month (Table I). Sampling uses pickup date, hour and five exclusive corridor strata: airport, intra-Manhattan, Manhattan to outer boroughs, outer boroughs to Manhattan, and outer to outer. The construction uses minimum-one adjusted proportional quotas and SHA-256 ranking. October diagnostic and November–December test trips are stratified by date and hour. This cohort is a sample of the source data and does not include all eligible NYC trips. Seven month-balanced subsets retain increasing prefixes of the hash-ranked monthly pool:

$$S_{90k} \subset S_{180k} \subset S_{360k} \subset S_{540k} \subset S_{720k} \subset S_{900k} \subset S_{1080k}. \quad (1)$$

Their monthly sizes are 10,000, 20,000, 40,000, 60,000, 80,000, 100,000 and 120,000. Each larger tier retains every trip in the smaller tiers, so comparisons do not replace one sample with another. All tiers share the same 20,000 test trips. Feature ablations and reliance diagnostics use S90k models. History density and neural update count also increase with tier size.

B. Feature sets and tier-isolated history

F1 and F2 each extend F0; F3 is their union of 34 features (Table II). Pickup and drop-off zone identifiers are derived from coordinates by a spatial join with TLC taxi-zone polygons. The seven zone/geometry features are the two zone identifiers, Haversine distance, same-zone and airport flags, and encoded pickup/drop-off boroughs. Seven OSRM features describe distance, duration, maneuver steps, graph nodes and highway/primary/local distance fractions, using a static OSM graph dated January 1, 2015. Endpoints snapping more than 1 km from the network mark a route missing. There are no missing routes in S90k, diagnostic or test splits, with 299, 31 and 63 zero-distance routes respectively; the full training pool has 13 missing routes. The ten calendar features are hour, weekday, month, weekend and holiday flags, sine/cosine pairs for hour and weekday, and a 168-hour weekly slot. The three history encodings use OD, OD–hour and OD–slot cells. OD is the pickup/drop-off zone pair, so F2 indirectly carries spatial information. At pickup time t, history uses only training trips completed strictly before t:

$$\mathrm{TE}_g(t) = \frac{\sum_{j \in H_g(t)} z_j + m\bar{z}(t)}{|H_g(t)| + m}, \qquad m = 20. \quad (2)$$

Here $z_j = \log(1 + y_j)$ for duration y_j in minutes; $\bar{z}(t)$ is the mean log-duration of prior training completions, initialized to $\log(1 + 15)$ before the first completion. Sparse cells shrink toward that running mean. All history statistics are recomputed independently within each training tier. Diagnostic and test labels never update them; evaluation history remains frozen to January–September completions.

Table II
FEATURE SETS USED FOR REFITTED ABLATION.

Set	Count	Inputs
F0	7	Vendor, passenger count, four endpoint coordinates, pickup timestamp.
F1	21	F0 plus seven zone/geometry and seven OSRM features.
F2	20	F0 plus ten calendar and three history features.
F3	34	Base, spatial/routing and time/history features ($7 + 14 + 13$).

IV. EXPERIMENTAL PROTOCOL

A. Preprocessing, learners and configuration transfer

Fitted preprocessing uses only the active training partition. Missing numerical values use fit-partition medians. LightGBM and CatBoost receive float32 numerical arrays; CatBoost runs on four CPU threads without native categorical mode. Neural continuous inputs are standardized, while vendor, zone and borough columns use categorical representations. RealMLP maps unseen categories to the training mode; TabM reserves index zero for them.

All learners minimize squared error on log-duration in minutes (RMSE for CatBoost). Predictions are clipped to $[-20, 12]$ in log space, inverted as $\hat{y} = \max(0, \exp(\hat{z}) - 1)$ and converted to seconds. TabM standardizes the target with the fit-partition mean and population SD and averages its 32 heads before inversion. Both neural learners use three hidden blocks of width 256 and batch size 512. LightGBM uses column sampling 0.85 and no row bagging.

Hyperparameters were selected on F3 using Optuna 4.5.0 TPE [18], proposal seed 20261026 and model seed 42. The inner split contains 79,997 January–August trips completed before September 1, and 10,000 September validation trips, scored by MAE in seconds. The completion cutoff excludes three of the nominal 80,000 January–August trips. Search budgets are 10/15/10/10 trials for LightGBM/CatBoost/RealMLP/TabM. Selected configurations and training budgets are retained across refits (Table III). Final refits do not use validation-based early stopping: RealMLP completes 24 epochs within its 120-epoch schedule, and TabM completes 39 epochs. Neither diagnostic nor test scores determine when these refits stop. The factorial design comprises $4 \times 4 \times 3 = 48$ fits. Scaling comprises $7 \times 4 \times 3 = 84$ learner evaluations and $7 \times 2 = 14$ deterministic reference evaluations, or 98 evaluations in total. The 12 S90k/F3 learner evaluations are shared between the factorial and scaling analyses and are not additional independent fits.

Table III
LOCKED CONFIGURATIONS FOR FINAL REFITS. TREE AND EPOCH COUNTS ARE FIXED; PATIENCE IS NOT USED DURING THESE REFITS.

Learner	Configuration
LightGBM	2,500 trees; learning rate 0.01239; depth 9; 96 leaves; minimum child samples 10; L1 4.334; L2 0; column sample 0.85.
CatBoost	1,307 trees; learning rate 0.03463; depth 9; L2 0; random strength 0; four CPU threads.
RealMLP	24 fixed epochs; schedule horizon 120; learning rate 0.1415; weight decay 0.001834; dropout 0.2628.
TabM	39 fixed epochs; learning rate 0.0003307; weight decay 0; dropout 0.1106; 32 heads.

B. Metrics and deterministic references

We report means and sample SDs over seeds 42–44 on 20,000 test trips. Metrics are MAE, RMSE, median absolute error (MedAE), WAPE (total absolute error divided by total observed duration), R^2, and RMSLE computed with $\log(1+y)$ in minutes. Duration errors are reported in seconds.

The uncalibrated OSRM reference uses positive static route duration, with fallback training medians by OD pair, weekly slot and globally. It does not model measured traffic. TEMP estimates mean Haversine speed per weekly slot, multiplies each historical duration by its slot speed, averages within OD pairs and divides by the query slot speed, with the same fallback hierarchy. Both references use the corresponding training tier. We do not evaluate a calibrated OSRM baseline.

C. Inference and reliance diagnostics

The primary feature contrast is $\Delta_m = \mathrm{MAE}_{m,F1} - \mathrm{MAE}_{m,F2}$; positive values favor F2. Scaling contrasts compare S720k with S90k within a learner, or RealMLP with other learners at S720k. We use 5,000 paired pickup-day bootstrap resamples of the 61 test days (seed 20261026), aggregating daily error sums and trip counts after averaging absolute errors over model seeds. Pointwise intervals use the 2.5th and 97.5th percentiles; Bonferroni-adjusted intervals use tails $0.025/K$ and $1 - 0.025/K$. Families are $K = 4$ for F1–F2 and within-learner scaling, $K = 5$ for S90k/F3 comparisons against RealMLP, and $K = 3$ for cross-learner comparisons at S720k. These families do not account for selecting S720k after comparing tiers.

Grouped permutation uses all twelve S90k/F3 checkpoints and 10,000 October trips. Groups are raw coordinates (4), zones/geometry (7), OSRM routes (7), calendar (10) and history (3). Vendor, passenger count and raw pickup timestamp remain fixed. We use ten shared permutations (seed 20261023) and 5,000 bootstrap resamples of the 31 diagnostic days (seed 20261024), with $K = 20$ across four learners and five groups. Antithetic permutation-SHAP estimates use each learner's seed-42 checkpoint, 64 randomly sampled October trips and 32 training background trips (seed 42), with $4 \times 34 + 2 = 138$ evaluations per case. This is an approximation, not exact Shapley evaluation.

Table IV
S90K TEST PERFORMANCE ACROSS FEATURE SETS: MEAN (SAMPLE SD) OVER SEEDS 42–44.

Model	F0	F1	F2	F3
Panel A: MAE (seconds)				
LightGBM	261.17 (0.65)	252.15 (0.10)	216.32 (0.71)	203.39 (0.68)
CatBoost	264.08 (0.34)	253.79 (0.30)	213.36 (0.39)	203.21 (0.36)
RealMLP	253.67 (0.50)	252.11 (0.81)	204.17 (0.63)	200.14 (1.18)
TabM	257.04 (1.30)	250.48 (0.16)	216.75 (4.17)	204.90 (0.85)
Panel B: RMSE (seconds)				
LightGBM	409.03 (0.53)	399.87 (0.18)	347.59 (0.48)	333.59 (0.55)
CatBoost	412.17 (0.20)	403.00 (0.50)	346.63 (1.01)	333.96 (0.44)
RealMLP	416.93 (1.68)	404.29 (1.12)	347.61 (2.88)	333.96 (0.30)
TabM	433.83 (5.91)	418.60 (1.92)	380.63 (11.63)	354.31 (3.61)
Panel C: R^2				
LightGBM	0.6441 (0.0009)	0.6599 (0.0003)	0.7430 (0.0007)	0.7633 (0.0008)
CatBoost	0.6386 (0.0004)	0.6545 (0.0009)	0.7444 (0.0015)	0.7628 (0.0006)
RealMLP	0.6302 (0.0030)	0.6523 (0.0019)	0.7430 (0.0042)	0.7628 (0.0004)
TabM	0.5996 (0.0109)	0.6273 (0.0034)	0.6916 (0.0190)	0.7330 (0.0054)

Table V
PAIRED S90K TEST MAE CONTRAST $\mathrm{MAE}_{F1} - \mathrm{MAE}_{F2}$ (SECONDS): 5,000 RESAMPLES OF 61 TEST DAYS; BONFERRONI $K = 4$.

Learner	Difference	Pointwise 95% CI	Adjusted 95% CI
LightGBM	35.83	[29.03,41.87]	[27.25,43.57]
CatBoost	40.42	[34.13,46.26]	[32.40,47.94]
RealMLP	47.95	[41.77,53.79]	[39.64,55.20]
TabM	33.73	[28.07,39.22]	[26.42,40.50]

V. EXPERIMENTAL RESULTS

A. Feature ablation on S90k

Table IV reports MAE, RMSE and R^2 across feature sets. F3 gives the lowest mean MAE and RMSE and highest mean R^2 for each learner. F1 reaches 250.48–253.79 s MAE, compared with TEMP at 255.97 s; F2 reaches 204.17–216.75 s. The direct F1–F2 contrasts are 35.83, 40.42, 47.95 and 33.73 s for LightGBM, CatBoost, RealMLP and TabM, respectively; all adjusted intervals exclude zero (Table V). Adding spatial/routing features lowers mean MAE by 1.56–10.29 s from F0 to F1 and 4.03–12.93 s from F2 to F3. Adding time/history lowers it by 40.29–50.71 s from F0 to F2 and 45.58–51.97 s from F1 to F3 (Fig. 1). The intervals in the feature-contrast table compare F1 with F2; they do not test each transition shown in the figure.

Table VI gives all six metrics for F3 and the references. RealMLP has the lowest mean MAE, WAPE and RMSLE; LightGBM the lowest RMSE and highest R^2; TabM the lowest median error but heavier tails. RealMLP-minus-reference adjusted intervals exclude zero for OSRM (-327.17 s, $[-353.75, -297.68]$) and TEMP (-55.83 s, $[-63.03, -47.68]$). Its contrasts with CatBoost (-3.07 s, $[-6.91, 1.25]$), LightGBM (-3.25 s, $[-7.04, 0.53]$) and TabM (-4.76 s, $[-10.51, 2.38]$) do not exclude zero. Only comparisons against RealMLP were tested, and intervals containing zero do not establish equivalence.

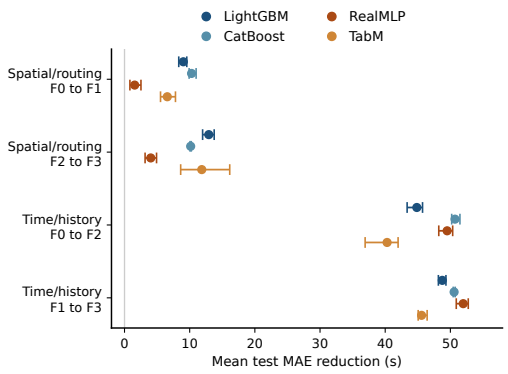


Fig. 1. Mean test MAE reductions across seeds 42–44 on S90k. Horizontal intervals show the minimum and maximum paired seed reductions; they are not confidence intervals.

Table VI
FULL-FEATURE S90K TEST BENCHMARK; PARENTHESES GIVE SAMPLE SD ACROSS THREE SEEDS.

Pipeline	MAE	RMSE	MedAE
Panel A: duration errors (seconds)			
OSRM	527.31	733.35	384.70
TEMP	255.97	424.80	160.58
LightGBM	203.39 (0.68)	333.59 (0.55)	126.99 (0.48)
CatBoost	203.21 (0.36)	333.96 (0.44)	127.45 (0.44)
RealMLP	200.14 (1.18)	333.96 (0.30)	122.81 (0.72)
TabM	204.90 (0.85)	354.31 (3.61)	118.16 (0.57)
Panel B: dimensionless metrics			
Pipeline	WAPE	RMSLE	R^2
OSRM	0.6125	0.8879	−0.1440
TEMP	0.2973	0.3933	0.6161
LightGBM	0.2363 (0.0008)	0.3076 (0.0008)	0.7633 (0.0008)
CatBoost	0.2361 (0.0004)	0.3048 (0.0005)	0.7628 (0.0006)
RealMLP	0.2325 (0.0014)	0.3027 (0.0008)	0.7628 (0.0004)
TabM	0.2380 (0.0010)	0.3142 (0.0008)	0.7330 (0.0054)

B. Fitted-model reliance on October trips

OSRM permutation yields the largest MAE increase for every learner, 252.02–301.36 s (Table VII). Calendar penalties are 84.60–100.12 s; history penalties are 4.34–19.44 s, with all four adjusted history intervals excluding zero. TabM has a small coordinate penalty (5.58 s) and a larger zones/geometry penalty (67.22 s). This pattern differs from RealMLP, whose coordinate and zones/geometry penalties are 42.49 and 60.87 s. We did not test which learned representations account for this difference. OSRM distance and duration occupy the top two SHAP positions for LightGBM, CatBoost and RealMLP; OSRM duration ranks first for TabM (Fig. 2). Other top-eight features include Haversine distance, cyclic time encodings, raw timestamp, OD history and zone identifiers. We report the lower ranks descriptively: the SHAP estimates use few permutations, and their uncertainty was not quantified.

Table VII
OCTOBER GROUPED-PERMUTATION MAE INCREASES (SECONDS), S90K/F3 MODELS, 10,000 TRIPS. THE FINAL ROW REPORTS BONFERRONI-ADJUSTED 95% INTERVALS FOR HISTORY ONLY ($K = 20$ ACROSS ALL GROUPS AND LEARNERS); INTERVALS FOR THE OTHER GROUPS ARE OMITTED FROM THIS TABLE.

Group	LightGBM	CatBoost	RealMLP	TabM
Coordinates	36.81	41.72	42.49	5.58
Zones/geometry	24.37	33.14	60.87	67.22
OSRM routes	301.36	269.47	252.02	271.72
Calendar	97.53	95.03	100.12	84.60
History	6.03	19.44	4.34	10.42
Adjusted 95% CI	[3.76,8.66]	[13.04,26.43]	[2.13,6.70]	[7.72,13.52]

C. Nested sample-scaling analysis

Table VIII and Fig. 3 summarize the seven tiers. TEMP improves from 255.97 to 234.53 s as history becomes denser, while the uncalibrated OSRM reference changes little (527.31–527.38 s). Tree MAE remains within 199.50–203.39 s under the transferred configurations. At S720k, LightGBM changes by -0.69 s relative to S90k (adjusted interval $[-2.33, 1.09]$), and CatBoost by -3.72 s ($[-5.25, -2.06]$; Table IX). RealMLP improves to 195.14 ± 0.24 s at S360k, 192.72 ± 0.70 s at S720k and 192.97 ± 1.18 s at S900k. The lowest observed mean occurs at S720k. The nearby tier result and variation across seeds give little basis for identifying a unique optimum or a transition at S360k. The S720k-minus-S90k contrast is -7.42 s ($[-11.54, -3.85]$ adjusted), compared with -2.29 s ($[-8.20, 4.67]$) for TabM. At S720k, RealMLP-minus-competitor adjusted intervals exclude zero for LightGBM (-9.98 s), CatBoost (-6.78 s) and TabM (-9.88 s). This tier was selected after comparing test scores, so these intervals should be read as exploratory comparisons of the fitted models. With batch size 512 and fixed 24/39 epochs, increasing the sample 12-fold also increases update counts from 4,219 to 50,625 for RealMLP and from 6,856 to 82,266 for TabM. At S900k and S1080k, TabM reaches 208.64 ± 0.61 and 209.87 ± 3.11 s. RealMLP reaches 200.08 ± 12.19 s at S1080k: seeds 42, 43 and 44 give 192.34, 214.12 and 193.76 s respectively. The higher mean is driven by one seed. These final scores do not show when or why that run deteriorated.

VI. DISCUSSION

A. Refitted gains and fitted-model reliance

Permuting OSRM descriptors raises MAE by 252.02–301.36 s in S90k/F3 models, whereas removing the whole spatial/routing bundle and refitting (F3 to F2) costs 4.03–12.93 s on average. The measurements also differ in scope and period: permutation changes the OSRM group on October diagnostic trips, whereas refitting removes the full spatial/routing bundle and is evaluated on November–December test trips. Permuted routes can conflict with unchanged coordinates and derived features. Meanwhile, F2 may retain some of the same information through coordinates and history. Either effect could contribute to the gap; this experiment does not separate them. The

Table VIII
NESTED F3 SAMPLE-SCALING TEST MAE (SECONDS): MEAN ± SAMPLE SD FOR LEARNERS AND DETERMINISTIC REFERENCE SCORES. THE 98 EVALUATIONS COMPRISE 84 LEARNER AND 14 REFERENCE EVALUATIONS; 12 LEARNER EVALUATIONS ARE SHARED WITH THE FACTORIAL ANALYSIS.

Tier	Trips	OSRM	TEMP	LightGBM	CatBoost	RealMLP	TabM
S_{90k}	90,000	527.31	255.97	203.39 ± 0.68	203.21 ± 0.36	200.14 ± 1.18	204.90 ± 0.85
S_{180k}	180,000	527.36	248.39	202.36 ± 0.57	201.28 ± 0.14	198.81 ± 3.17	204.82 ± 3.39
S_{360k}	360,000	527.36	241.08	201.58 ± 0.50	199.93 ± 0.84	195.14 ± 0.24	200.70 ± 1.12
S_{540k}	540,000	527.34	238.15	202.63 ± 0.28	199.71 ± 0.52	194.99 ± 2.60	202.01 ± 3.41
S_{720k}	720,000	527.37	236.68	202.70 ± 0.71	199.50 ± 0.30	192.72 ± 0.70	202.61 ± 5.07
S_{900k}	900,000	527.38	235.68	202.43 ± 0.13	199.72 ± 0.26	192.97 ± 1.18	208.64 ± 0.61
S_{1080k}	1,080,000	527.37	234.53	202.81 ± 0.62	199.59 ± 0.65	200.08 ± 12.19	209.87 ± 3.11

Table IX
PAIRED SCALING CONTRASTS AND 95% INTERVALS: 5,000 RESAMPLES OF 61 TEST DAYS. ADJUSTMENTS COVER THE STATED COMPARISON FAMILIES, NOT TIER SELECTION.

Comparison	Difference	Pointwise CI	Adjusted CI
Panel A: S720k minus S90k ($K = 4$)			
LightGBM	-0.69	$[-2.02, 0.69]$	$[-2.33, 1.09]$
CatBoost	-3.72	$[-4.95, -2.40]$	$[-5.25, -2.06]$
RealMLP	-7.42	$[-10.64, -4.48]$	$[-11.54, -3.85]$
TabM	-2.29	$[-7.24, 3.21]$	$[-8.20, 4.67]$
Panel B: RealMLP minus competitor at S720k ($K = 3$)			
LightGBM	-9.98	$[-15.03, -4.90]$	$[-16.01, -3.71]$
CatBoost	-6.78	$[-11.28, -2.15]$	$[-12.18, -1.17]$
TabM	-9.88	$[-13.31, -6.57]$	$[-14.10, -5.99]$

permutation penalty therefore cannot estimate the gain from acquiring route features.

Within F3, calendar permutation causes a larger penalty than history permutation. Separate refits with calendar-only and history-only additions would be needed to determine how much each contributes to F2. OD history appears among the top eight SHAP features for LightGBM and CatBoost although its group permutation penalty is small. Correlation may distribute attribution across related features. Similarly, coordinate permutation leaves zones, Haversine distance and OSRM routes intact, which can reduce the apparent coordinate penalty.

B. Interpreting sample scaling

RealMLP's mean error is lower at S360k–S900k than at S90k while tree errors change less under the retained configurations. Larger tiers provide more trips, denser history encodings and more neural updates at the same time. Any of these changes could contribute to the improvement. The tree results may also depend on the fixed tree counts and limited HPO. The trajectories thus show how the selected configurations transfer across sample sizes; they do not establish a scaling law or a capacity limit for trees.

The S1080k RealMLP seed spread and TabM's larger-tier errors show that transferred configurations need not improve monotonically. Whether a different batch size, epoch count or learning-rate schedule would help remains untested. Extending an epoch-based schedule alone would not hold update count fixed: updates also depend on sample count, batch size and completed epochs. RealMLP also uses only the first 24 epochs of its 120-epoch schedule. The effects of that choice have not been tested separately.

C. Validity check

A post hoc S90k check was run only for LightGBM at seed 42. Omitting month gives 216.73 s MAE for F2, compared with 216.06 s with month and 252.24 s for F1; F3 without month gives 203.28 s versus 203.00 s with month. F2 retains its advantage in this LightGBM check. Its scope is limited to that learner and tier; the raw timestamp predictor was retained.

D. Limitations

Selecting hyperparameters on F3 at S90k may favor F3 over F0 and the other reduced feature sets, and reusing them at larger tiers may disadvantage configurations that need retuning. HPO budgets also differ across learners. LightGBM's selected count equals its 2,500-tree cap. We use one sampled cohort, one sampling seed and three model seeds. The test period was inspected during development, and S720k was highlighted after inspecting test scores. Tier-level conclusions are exploratory. Bootstrap intervals resample days while holding the fitted models fixed. They omit variation from refitting and cohort sampling, as well as uncertainty from selecting a tier after seeing its test score.

SHAP uses only 64 diagnostic trips and one checkpoint per learner, with a small permutation budget and no attribution intervals. Correlated route and geometry predictors can share credit. Permutation can create implausible inputs; keeping raw timestamp fixed may understate calendar reliance. Both timestamp and test months lie beyond the training time range. The month check covers LightGBM only and does not address timestamp extrapolation. Specific holidays differ between training and test periods, and aggregate errors may hide those differences. Errors were not analyzed separately by month or holiday.

Calendar and history are not separated by refitted ablations. Sparse OD–time cells, especially at S90k, may understate history's value relative to a full operational trip

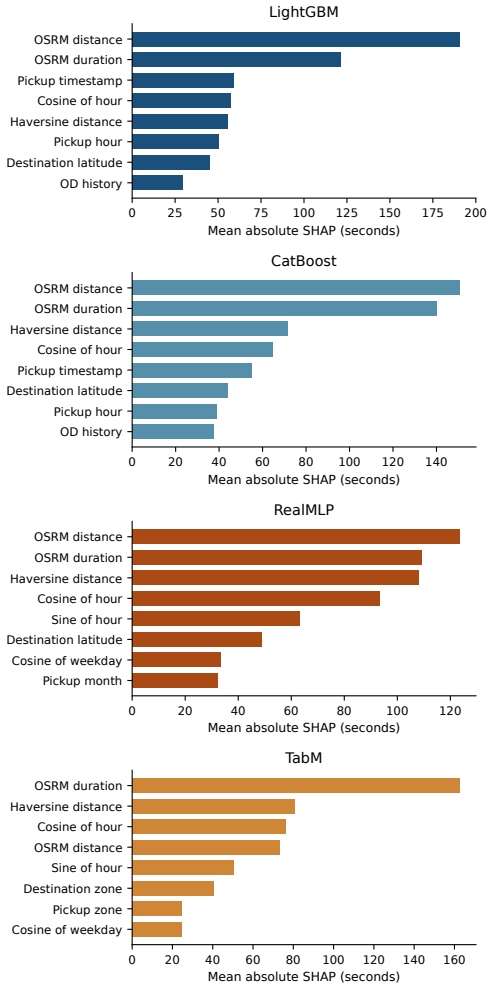


Fig. 2. Top eight mean absolute permutation-SHAP estimates per S90k/F3 learner (seed 42), using 64 October diagnostic trips. Values are in seconds.

stream. History density changes with tier size. October is excluded from final training, leaving a one-month gap before test, and test history is frozen to September completions. Eligibility uses realized duration and distance, recorded drop-off coordinates proxy intended destinations, and the static 2015 OSM graph omits live traffic and later road changes. OSRM is uncalibrated here, so its error does not represent a traffic-calibrated routing reference. There is also a mismatch between the squared log-error training objective and MAE or RMSE in seconds; the inverse transform applies no bias correction.

E. Data and code availability

The raw-data source is documented by Microsoft Azure Open Datasets [17]. The experimental code is maintained in the authors' project package. No public repository or permanent archive is linked in this paper.

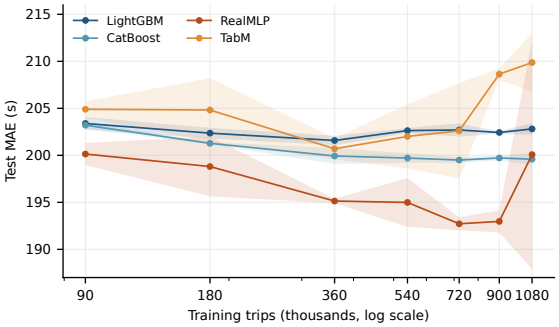


Fig. 3. Test MAE across nested F3 training tiers. Lines show three-seed means and shading shows one sample SD. Sample count, neural update count and history density vary together.

VII. Conclusion

On this chronological 2015 NYC taxi cohort, the combined time/history bundle gives 33.73–47.95 s lower test MAE than spatial/routing features at S90k, and F3 gives the lowest mean MAE for every learner. Permuting OSRM descriptors causes the largest error increase in the fitted full-feature models. This gap between refitted gains and perturbation penalties shows why both measurements are needed. The F2 result applies to calendar and history together; history's contribution remains unresolved.

Under transferred configurations, RealMLP reaches its lowest observed mean at S720k (192.72 ± 0.70 s) and remains close at S900k (192.97 ± 1.18 s). At S720k, adjusted intervals against the other learners exclude zero for these fitted models; tier selection was exploratory. At S1080k, one of three RealMLP seeds has substantially higher error. Further experiments would be needed to determine whether changing the update budget or schedule improves this result.

References

[1] Q. Au, J. Herbinger, C. Stachl, B. Bischl, and G. Casalicchio, "Grouped feature importance and combined features effect plot," *Data Min. Knowl. Discov.*, vol. 36, no. 4, pp. 1401–1450, 2022, doi: 10.1007/s10618-022-00840-5.

[2] A. Fisher, C. Rudin, and F. Dominici, "All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously," *J. Mach. Learn. Res.*, vol. 20, no. 177, pp. 1–81, 2019. [Online]. Available: https://jmlr.org/papers/v20/18-760.html

[3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 4765–4774. [Online]. Available: https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

[4] G. Hooker, L. Mentch, and S. Zhou, "Unrestricted permutation forces extrapolation: Variable importance requires at least one more model, or there is no free variable importance," *Stat. Comput.*, vol. 31, Art. no. 82, 2021, doi: 10.1007/s11222-021-10057-z.

[5] H. Wang, X. Tang, Y.-H. Kuo, D. Kifer, and Z. Li, "A simple baseline for travel time estimation using large-scale trip data," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, Art. no. 19, pp. 1–22, 2019, doi: 10.1145/3293317.

[6] D. Luxen and C. Vetter, "Real-time routing with OpenStreetMap data," in *Proc. 19th ACM SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, 2011, pp. 513–516, doi: 10.1145/2093973.2094062.

[7] D. Wang, J. Zhang, W. Cao, J. Li, and Y. Zheng, "When will you arrive? Estimating travel time based on deep neural networks," in *Proc. 32nd AAAI Conf. Artif. Intell.*, vol. 32, no. 1, 2018, pp. 2500–2507, doi: 10.1609/aaai.v32i1.11877.

[8] H. Huang, M. Pouls, A. Meyer, and M. Pauly, "Travel time prediction using tree-based ensembles," arXiv:2005.13818, 2020. [Online]. Available: https://arxiv.org/abs/2005.13818

[9] S. Schleibaum, J. P. Müller, and M. Sester, "An explainable stacked ensemble model for static route-free estimation of time of arrival," *J. Adv. Transp.*, vol. 2024, Art. no. 9301691, 2024, doi: 10.1155/2024/9301691.

[10] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017. [Online]. Available: https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

[11] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased boosting with categorical features," in *Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 6639–6649. [Online]. Available: https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html

[12] D. Holzmüller, L. Grinsztajn, and I. Steinwart, "Better by default: Strong pre-tuned MLPs and boosted trees on tabular data," in *Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, doi: 10.52202/079017-0837.

[13] Y. Gorishniy, A. Kotelnikov, and A. Babenko, "TabM: Advancing tabular deep learning with parameter-efficient ensembling," in *Proc. Int. Conf. Learn. Representations*, 2025. [Online]. Available: https://openreview.net/forum?id=Sd4wYYOhmY

[14] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, doi: 10.52202/068431-0037.

[15] D. McElfresh et al., "When do neural nets outperform boosted trees on tabular data?" in *Adv. Neural Inf. Process. Syst.*, vol. 36, 2023. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/f06d5ebd4ff40b40dd97e30cee632123

[16] V. Cherepanova et al., "A performance-driven benchmark for feature selection in tabular deep learning," in *Adv. Neural Inf. Process. Syst.*, vol. 36, 2023. https://arxiv.org/abs/2311.05877

[17] Microsoft Azure Open Datasets, "NYC TLC Yellow Taxi trip records: Source container," 2015 partitions, accessed Oct. 2026. [Online]. Available: https://learn.microsoft.com/en-us/azure/open-datasets/dataset-taxi-yellow

[18] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. 25th ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2019, pp. 2623–2631, doi: 10.1145/3292500.3330701.